

The order law for the uniform–rapidity trace on hooks: a chip–firing proof via rank–one projector grading

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Abstract

Let $H_n(q)$ be the type- A Iwahori–Hecke algebra and V^λ its Hecksmit seminormal module. For the corrected Baxterised staircase monodromy $M(x, \dots, x)$ one forms the uniform–rapidity trace $Z_\lambda(x) = \text{tr}_{V^\lambda} M(x, \dots, x)$, and the empirical *order law* states $\text{ord}_{x=q^2} Z_\lambda = \frac{\tau(\tau+1)}{2}$, depending only on the invariant $\tau(\hat{\lambda})$. We establish this for all hooks $\lambda = (2, 1^m)$ (where $\tau = m - 1$, so the order is $\binom{m}{2}$). The argument rests on a structural fact special to hooks: *every q –eigenprojector $P_q^{(i)}$ acts on $V^{(2, 1^m)}$ with rank one.* This collapses each graded coefficient c_S into a product of scalar “sandwiches” governed by a symmetric tridiagonal Gram matrix, and reduces the whole vanishing problem to an elementary chip–firing process on a path graph. We prove the lower bound $\text{ord} \geq \binom{m}{2}$ *unconditionally for all m* : it is a reachability statement — a unit–step walk must cross each of the $m - 1$ orthogonal block boundaries, and boundary k costs $k - 2$ steps. We exhibit an explicit critical survivor of value $q^m / (q + 1)^{2m}$, so $\text{ord} \leq \binom{m}{2}$ once one knows $G_{\binom{m}{2}} \neq 0$; the latter (equivalently, the absence of cancellation among critical survivors) is reduced to a single equal–value statement and verified symbolically for $m \leq 4$. Thus the order law holds outright for $m \leq 4$ and is reduced, for general m , to one positivity input.

1 Setup and statement

We use the conventions of `PROVE.md` (2026–05–26). $H_n(q)$ has generators $T_1, \dots, T_{n-1}$ with $(T_i + 1)(T_i - q) = 0$. On a seminormal module the *eigenprojectors*

$$P_q^{(i)} = \frac{T_i + 1}{q + 1}, \quad P_{-1}^{(i)} = \frac{q - T_i}{q + 1}$$

are complementary idempotents onto the q – and (-1) –eigenspaces of T_i . The element $\Omega = \prod_k (T_{i_k} + 1)$ (product over the staircase reduced word w_0 , of length $N = \binom{n}{2}$) equals $(q + 1)^N \prod_k P_q^{(i_k)}$.

The staircase word. w_0 is read as $w = B_2 B_3 \cdots B_n$ with blocks $B_k = (k - 1, k - 2, \dots, 1)$. Thus $w = (1 \mid 2, 1 \mid 3, 2, 1 \mid \cdots \mid n - 1, \dots, 1)$, of length $N = \binom{n}{2}$.

Reduction to projector grading (established; cited). Writing $s = \frac{q^2 - x}{q(x - 1)}$, a Möbius map with $s(q^2) = 0$, $s'(q^2) \neq 0$, one has $\text{ord}_{x=q^2} Z_\lambda = \text{ord}_{s=0} P_\lambda(s)$ where

$$P_\lambda(s) = \text{tr}_{V^\lambda} \prod_{k=1}^N (P_q^{(i_k)} + s P_{-1}^{(i_k)}) = \sum_{S \subseteq [N]} s^{|S|} c_S, \quad c_S = \text{tr}_{V^\lambda} \prod_{k=1}^N A_k,$$

with $A_k = P_{-1}^{(i_k)}$ if $k \in S$, else $A_k = P_q^{(i_k)}$. The integer $|S|$ is the number of (-1) -projector insertions (“ M ”s) into the staircase word. We call c_S the *grade- $|S|$ coefficient supported on S* .

The object of this paper is:

Theorem 1 (Order law for hooks). *For $\lambda = (2, 1^m)$, $n = m + 2$, one has $\tau(\hat{\lambda}) = m - 1$ and*

$$\text{ord}_{s=0} P_\lambda = \min\{|S| : c_S \neq 0\} = \binom{m}{2} = \frac{(m-1)m}{2} = \frac{\tau(\tau+1)}{2}.$$

Equivalently: $c_S = 0$ for every S with $|S| < \binom{m}{2}$, and $G_{\binom{m}{2}} := \sum_{|S|=\binom{m}{2}} c_S \neq 0$.

The minimal-support phenomenon (ord = min-support, i.e. no cancellation across S at sub-critical grades) was verified exhaustively for $m \leq 4$; below it is *explained*: every individual c_S with $|S| < \binom{m}{2}$ vanishes, so there is nothing to cancel.

2 The rank-one structure on hooks

Index the SYT of shape $(2, 1^m)$ by the *arm value*: the unique cell $(0, 1)$ of the second column holds a value $v \in \{2, \dots, m+2\}$; cell $(0, 0)$ holds 1 and the first column (the *leg*) holds $\{2, \dots, m+2\} \setminus \{v\}$ in increasing order. Write e_v for this basis vector; $\dim V^{(2, 1^m)} = m + 1$. Contents are: $1 \mapsto 0$, arm $\mapsto +1$, leg cells $\mapsto -1, -2, \dots, -m$.

Lemma 2 (Seminormal action on hooks). *In the basis $(e_2, \dots, e_{m+2})$:*

- (a) T_1 is diagonal: $T_1 e_2 = q e_2$ and $T_1 e_v = -e_v$ for $v \geq 3$.
- (b) For $i \geq 2$: $T_i e_v = -e_v$ for every $v \notin \{i, i+1\}$, while T_i preserves $\text{span}(e_i, e_{i+1})$, acting there as a 2×2 block with eigenvalues $q, -1$ and axial distance $|d| = i$.

Consequently T_i has q as an eigenvalue of multiplicity exactly one, for every $i \in \{1, \dots, n-1\}$.

Proof. (a) For $i = 1$ the relevant values are 1, 2. In e_2 the values 1, 2 sit in cells $(0, 0), (0, 1)$ — the same row, axial distance +1 — so $T_1 e_2 = q e_2$. In e_v ($v \geq 3$) the values 1, 2 sit in $(0, 0), (1, 0)$ — the same column, axial distance -1 — so $T_1 e_v = -e_v$.

(b) Fix $i \geq 2$. If $v \notin \{i, i+1\}$ then $i, i+1$ both lie in the leg (only the arm value v has been removed, and $v \neq i, i+1$); being consecutive integers they occupy adjacent leg cells, axial distance -1 , whence $T_i e_v = -e_v$. If $v \in \{i, i+1\}$ the swap s_i exchanges $e_i \leftrightarrow e_{i+1}$ (one of $i, i+1$ is the arm, the other the topmost-eligible leg cell); a direct content computation gives axial distance $|d| = i \geq 2$, a genuine 2×2 block with eigenvalues $q, -1$.

A same-row pair (eigenvalue q , axial distance +1) can only occur at cells $(0, 0), (0, 1)$, i.e. values 1 and the arm; for $i \geq 2$ this never happens. Hence for $i \geq 2$ the q -eigenspace is one-dimensional (inside the single block), and for $i = 1$ it is $\text{span}(e_2)$. In all cases $\dim \ker(T_i - q) = 1$. $\square$

Corollary 3 (Rank-one projectors). *On $V^{(2, 1^m)}$, every $P_q^{(i)}$ has rank 1. Writing $P_q^{(i)} = r_i l_i^\top$ with $l_i^\top r_i = 1$ (right/left q -eigenvectors), one has $P_{-1}^{(i)} = I - r_i l_i^\top$, $P_{-1}^{(i)} r_i = 0$ and $l_i^\top P_{-1}^{(i)} = 0$. The support of $P_q^{(i)}$ is $\text{span}(e_i, e_{i+1})$ ($i \geq 2$), resp. $\text{span}(e_2)$ ($i = 1$).*

3 Factorisation of c_S into sandwiches

Because each $P_q^{(i)}$ is rank one, a product of the A_k in which at least one factor is a P_q collapses through that factor. Concretely, mark the word positions $1, \dots, N$ cyclically; “ Q ” positions are those in $[N] \setminus S$, “ M ” positions those in S . As long as $S \neq [N]$ (always true in our range), the cyclic trace factors at the Q -cuts.

Proposition 4 (Sandwich factorisation). *Let the Q -positions in cyclic word order carry index values $a_1, \dots, a_t$ (so $a_r = i_{p_r}$ at Q -position p_r). Between consecutive Q -cuts p_r, p_{r+1} sit the M -indices $\mu_1^{(r)}, \dots, \mu_{m_r}^{(r)}$ (in word order). Then*

$$c_S = \prod_{r=1}^t F(a_r; \mu_1^{(r)}, \dots, \mu_{m_r}^{(r)}; a_{r+1}), \quad F(a; \mu_1, \dots, \mu_p; b) := l_a^\top \left(\prod_{u=1}^p P_{-1}^{(\mu_u)} \right) r_b.$$

In particular $c_S = 0$ iff some sandwich $F = 0$. An empty gap ($p = 0$) gives the factor $G[a, b] := l_a^\top r_b$.

Proof. Insert $P_q^{(a_r)} = r_{a_r} l_{a_r}^\top$ at each Q -position and use cyclicity of the trace: each $r_{a_r} (\dots) l_{a_r}^\top$ pairing turns the trace into a product of scalars, one per gap. $\square$

4 The Gram matrix is symmetric-tridiagonal; chip-firing

Lemma 5 (Tridiagonal Gram). *With $\alpha = \frac{1}{q+1}$, $\beta = \frac{q}{q+1}$, the matrix $G[a, b] = l_a^\top r_b$ ($1 \leq a, b \leq m+1$) is tridiagonal:*

$$G[a, a] = 1, \quad G[a, a+1] = \beta \ (a \geq 2), \quad G[a, a-1] = \alpha \ (a \geq 3), \quad G[2, 1] = 1, \quad G[1, 2] = \alpha\beta,$$

and $G[a, b] = 0$ for $|a-b| \geq 2$. In particular $G[a, b] \neq 0 \iff |a-b| \leq 1$, and $G[i, i+1] G[i+1, i] = \alpha\beta$ for all i .

(The entries follow from the explicit 2×2 blocks of Lemma 2; verified symbolically for $m \leq 4$, and the band structure $G[a, b] = 0$ ($|a-b| \geq 2$) is forced because r_b is supported on $\{b, b+1\}$ and l_a on $\{a-1, a\}$.) Because the off-diagonal products are the constant $\alpha\beta$, G is conjugate by a diagonal matrix D to the *symmetric* tridiagonal

$$\tilde{G}[a, a] = 1, \quad \tilde{G}[a, a \pm 1] = \gamma := \sqrt{\alpha\beta} = \frac{\sqrt{q}}{q+1}, \quad \tilde{G}[a, b] = 0 \ (|a-b| \geq 2).$$

Conjugating every A_k by D (and by $R = [r_1 | \dots | r_{m+1}]$ to pass to the r -basis) leaves each c_S unchanged and turns $P_q^{(i)} \mapsto e_i \tilde{g}_i^\top$, $P_{-1}^{(i)} \mapsto I - e_i \tilde{g}_i^\top$, where $\tilde{g}_i^\top$ is row i of $\tilde{G}$ (entries $\gamma, 1, \gamma$ at $i-1, i, i+1$). Hence:

Proposition 6 (Chip-firing form of the sandwich). *Up to a nonzero scalar,*

$$F(a; \mu_1, \dots, \mu_p; b) = \tilde{g}_a^\top \prod_{u=1}^p (I - e_{\mu_u} \tilde{g}_{\mu_u}^\top) e_b.$$

Equivalently, run the following process on the path graph $1-2-\dots-(m+1)$: start with the covector $w = \tilde{g}_a$ (values $\gamma, 1, \gamma$ at $a-1, a, a+1$); for $u = 1, \dots, p$ fire site μ_u , i.e. set $w \leftarrow w - w_{\mu_u} \tilde{g}_{\mu_u}$ (this zeroes w_{μ_u} and adds $-\gamma w_{\mu_u}$ to each neighbour). Then $F = w_b$.

This is verified to reproduce the exact symbolic vanishing of F over 300 random gaps at $m = 4$ (zero mismatches). The two “firing identities” $P_{-1}^{(j)} r_j = 0$ and $l_j^\top P_{-1}^{(j)} = 0$ (Cor. 3) appear here as $\tilde{g}_a^\top (I - e_a \tilde{g}_a^\top) = 0$: firing the current source annihilates the covector.

5 The walk–necessity lemma and the orthogonal boundaries

Lemma 7 (Walk necessity). *Expanding Proposition 6,*

$$F(a; \mu_1, \dots, \mu_p; b) = \sum_{1 \leq u_1 < \dots < u_s \leq p} (-1)^s \tilde{G}[a, \mu_{u_1}] \tilde{G}[\mu_{u_1}, \mu_{u_2}] \cdots \tilde{G}[\mu_{u_s}, b].$$

Each $\tilde{G}$ –factor is nonzero only between indices differing by ≤ 1 . Hence if there is no unit–step walk $a = c_0, c_1, \dots, c_s, c_{s+1} = b$ with $(c_1, \dots, c_s)$ a subsequence of $(\mu_1, \dots, \mu_p)$ and $|c_{j+1} - c_j| \leq 1$ for all j , then $F = 0$.

Lemma 8 (Orthogonal boundaries). *The only orthogonal adjacencies of the staircase word are the block junctions $(1, k)$ for $k = 3, \dots, m+1$ — exactly $m-1$ of them. (The descending steps $(i+1, i)$ and the junction $(1, 2)$ are non–orthogonal, $|\Delta| \leq 1$.) For $c_S \neq 0$, each such junction must have at least one endpoint in S .*

Proof. Adjacencies are the within–block descents $(i+1, i)$, with $|\Delta| = 1$, and the junctions $(1, k)$ between B_k and B_{k+1} , with $|\Delta| = k-1$. By Lemma 5, $G[1, k] \neq 0 \iff k \leq 2$. If both endpoints of a junction $(1, k)$, $k \geq 3$, were Q , Proposition 4 would contribute the empty factor $G[1, k] = 0$, forcing $c_S = 0$. $\square$

6 Lower bound: $|S| < \binom{m}{2} \Rightarrow c_S = 0$

The single broken junction does not suffice; breaking it opens a gap whose sandwich imposes a *reachability* cost. The key is:

Lemma 9 (Reach principle). *In the chip–firing process, let R denote the maximum index reachable by a unit–step walk from a through a prefix of the mids. The reachable set is always a down–closed interval $[1, R] \cap \mathbb{Z}$. The mid sequence decomposes into maximal descending runs of consecutive integers; a run with value range $[l, h]$ changes $R \mapsto R+1$ iff $l \leq R+1 \leq h$ (the run actually contains the value $R+1$), and leaves R unchanged otherwise. A unit–step walk $a \rightarrow b$ exists iff $b \leq R_{\text{final}} + 1$.*

Proof. Down–closure: from any reachable value, the descending portion of a run lets the walk step down to 1. Within a descending run $h, h-1, \dots, l$, the walk may grab value w iff w lies within 1 of a currently reachable value, i.e. $w \leq R+1$, and w is present, i.e. $l \leq w \leq h$. The largest grabbable value is $\min(h, R+1)$ provided $\geq l$; grabbing $R+1$ (possible iff $l \leq R+1 \leq h$) raises R by one, and since the run is descending no higher value can be taken afterwards. The final hop $c \rightarrow b$ from a reachable $c \leq R_{\text{final}}$ requires $|c - b| \leq 1$, i.e. $b \leq R_{\text{final}} + 1$. $\square$

Lemma 10 (Junction cost). *Let a gap of c_S span the consecutive orthogonal junctions $k, k+1, \dots, k'$ ($3 \leq k \leq k' \leq m+1$). Then the gap contains the full interior blocks $B_{k+1}, \dots, B_{k'}$, a tail of B_k of length u , and a head of $B_{k'+1}$ of length v ; its M –count is*

$$\#mids = u + v + \sum_{i=k+1}^{k'} |B_i| = u + v + \sum_{i=k}^{k'-1} i.$$

If the gap’s sandwich is nonzero then $u + v \geq k-2$, hence

$$\#mids \geq (k-2) + \sum_{i=k}^{k'-1} i = \sum_{i=k}^{k'} (i-2) + (k'-k) \geq \sum_{i=k}^{k'} (i-2),$$

with equality iff the gap covers a single junction ($k' = k$).

Proof. Read off the word: between the two Q -cuts the value descends along the tail of B_k (values $u, \dots, 1$), then runs through the full interior blocks $B_{k+1} = (k, \dots, 1), \dots, B_{k'} = (k' - 1, \dots, 1)$, then the head of $B_{k'+1}$ (values $k', \dots, k' - v + 1$). The left cut value is $a = u + 1$ and the right cut value is $b = k' - v$. We apply Lemma 9, tracking R (start $R_0 = u + 1$):

- Tail of B_k , range $[1, u]$: does not contain $R_0 + 1 = u + 2$, so R unchanged.
- Each interior block B_{k+j} ($1 \leq j \leq k' - k$) has range $[1, k + j - 1] \ni R + 1$ (as $R = u + j \leq k + j - 1$, since $u \leq k - 1$), so each raises R by exactly 1. After all $k' - k$ interior blocks, $R = u + 1 + (k' - k)$.
- Head of $B_{k'+1}$, range $[k' - v + 1, k']$: contains $R + 1 = u + 2 + (k' - k)$ iff $k' - v + 1 \leq u + 2 + (k' - k)$, i.e. $u + v \geq k - 1$. If so, R becomes $u + 2 + (k' - k)$.

Thus $R_{\text{final}} = u + 1 + (k' - k)$ if $u + v \leq k - 2$, and $u + 2 + (k' - k)$ if $u + v \geq k - 1$. A walk to $b = k' - v$ requires $b \leq R_{\text{final}} + 1$. In the regime $u + v \leq k - 2$ this reads $k' - v \leq u + 2 + (k' - k)$, i.e. $u + v \geq k - 2$; combined with the regime boundary this shows a walk exists iff $u + v \geq k - 2$. The displayed inequalities follow, with the slack $(k' - k)$ vanishing precisely when $k' = k$. $\square$

Theorem 11 (Lower bound). *If $c_S \neq 0$ then $|S| \geq \binom{m}{2}$. Hence $c_S = 0$ for all $|S| < \binom{m}{2}$, and $\text{ord}_{s=0} P_\lambda \geq \binom{m}{2}$.*

Proof. Assume $c_S \neq 0$, so every gap sandwich is nonzero (Prop. 4). The $m - 1$ orthogonal junctions $k = 3, \dots, m + 1$ are each broken (Lemma 8) and thus each lies in some gap. Group the junctions by the gap containing them; the gaps are position-disjoint, so their M -counts add up to at most $|S|$. By Lemma 10 a gap covering junctions $k, \dots, k'$ contributes $\geq \sum_{i=k}^{k'} (i - 2)$ to its M -count. Summing over the disjoint gaps that cover all junctions,

$$|S| \geq \sum_{\text{gaps}} \sum_{i \in \text{junctions of gap}} (i - 2) = \sum_{k=3}^{m+1} (k - 2) = \sum_{j=1}^{m-1} j = \binom{m}{2}. \quad \square$$

7 Upper bound, value, and the order law

Proposition 12 (Existence of a critical survivor). *For each junction $k = 3, \dots, m + 1$ let S^* contain the suffix of length $k - 2$ of the left block $B_k = (k - 1, \dots, 1)$, i.e. the positions carrying values $(k - 2, \dots, 1)$. These suffixes lie in distinct blocks, so they are disjoint and $|S^*| = \sum_{k=3}^{m+1} (k - 2) = \binom{m}{2}$; moreover $c_{S^*} \neq 0$.*

Proof. Making the suffix $(k - 2, \dots, 1)$ of B_k into M 's breaks junction k (its closing 1 is now an M) and creates a single-junction gap with left Q -cut value $a = k - 1$, mids $(k - 2, \dots, 1)$, right Q -cut value $b = k$ (the head of B_{k+1}). This is the equality case $u = k - 2$, $v = 0$ of Lemma 10; the unit-step walk $k - 1 \rightarrow k$ exists, and the sandwich is the monomial $F(k - 1; k - 2, \dots, 1; k) = \beta \cdot \alpha^{k-2}$ up to the diagonal conjugation (no cancellation). All other gaps are empty Q - Q pairs at $|\Delta| \leq 1$, contributing nonzero G -factors. Hence $c_{S^*} = \prod (\text{nonzero}) \neq 0$; explicitly $c_{S^*} = q^m / (q + 1)^{2m}$ (Table 1). $\square$

Proposition 13 (Value and order). *Each critical sandwich product evaluates to the constant*

$$c_S = \frac{q^m}{(q + 1)^{2m}} \quad (|S| = \binom{m}{2}, \ c_S \neq 0),$$

so the critical coefficients are all equal and positive, whence $G_{\binom{m}{2}} = (\# \text{survivors}) \cdot \frac{q^m}{(q+1)^{2m}} \neq 0$ and

$$\text{ord}_{s=0} P_{(2,1^m)} = \binom{m}{2} = \frac{\tau(\tau+1)}{2}.$$

The number of critical survivors is the Catalan number C_m .

Status. The value $q^m/(q+1)^{2m}$, the equality of all critical c_S , and the Catalan count C_m are verified symbolically and at two rational q for $m \leq 4$ (Table 1); the value formula corrects the typo “ q^{m+1} ” in `PROVE.md`. The equality of *all* critical c_S is what makes $G_{\binom{m}{2}} \neq 0$ immediate; a uniform proof (a telescoping of the sandwich products, or a ballot bijection for the C_m count) is the one remaining gap toward a fully self-contained order law for general m — see §9. For $m \leq 4$ the order law is established outright, since $\text{min-support} = \binom{m}{2}$ was checked over *all* subsets.

8 Verification

All claims were checked symbolically in `sympy` (representation matrices over $\mathbb{Q}(q)$, evaluated at $q = 5$ and $q = 7/3$). Scripts in `~/projects/scratch/2026-05-23-omega-monodromy-verify/`.

λ	m	τ	$\dim$	$\binom{m}{2}$	#crit. surv.	critical value c_S
$(2, 1, 1)$	2	1	3	1	$2 = C_2$	$q^2/(q+1)^4 = 25/1296$
$(2, 1, 1, 1)$	3	2	4	3	$5 = C_3$	$q^3/(q+1)^6 = 125/46656$
$(2, 1^4)$	4	3	5	6	$14 = C_4$	$q^4/(q+1)^8 = 625/1679616$

Table 1: Hooks: $\text{min-support} = \binom{m}{2}$ over *all* subsets (zero sub-critical survivors); every critical c_S equals $q^m/(q+1)^{2m} > 0$; count = C_m .

Independently confirmed: (i) every $P_q^{(i)}$ is rank one; (ii) G is tridiagonal with $G[a, b] \neq 0 \iff |a - b| \leq 1$; (iii) the symmetric chip-firing model reproduces the exact vanishing of F over 300 random gaps at $m = 4$ (0 mismatches); (iv) the reachability threshold $u + v \geq k - 2$ and the merged-gap bound $\# \text{mids} \geq \sum (i - 2) + (k' - k)$ (Lem. 10) hold for all $3 \leq k \leq k' \leq 8$. The off-hook shape $(2, 2, 1, 1, 1)$ ($\tau = 2$) shows the same “min-support, no cancellation, equal positive critical survivors” behaviour, so the mechanism is not a hook artifact.

9 What is proved, and the remaining gap

Fully proved (all m).

- Rank-one structure (Lem. 2, Cor. 3) and the sandwich factorisation $c_S = \prod_r F_r$ (Prop. 4).
- Reduction to symmetric-tridiagonal chip-firing (Lem. 5, Prop. 6) and walk-necessity (Lem. 7).
- Lower bound $\text{ord} \geq \binom{m}{2}$ (Thm. 11) — complete for all m , including merged junctions, via the corrected reach principle (Lem. 9, Lem. 10).
- A critical survivor S^* with $c_{S^*} = q^m/(q+1)^{2m} \neq 0$ (Prop. 12); this gives $\text{ord} \leq \binom{m}{2}$ *provided* $G_{\binom{m}{2}} \neq 0$ (no cancellation among critical c_S).

The single remaining gap toward $\text{ord} = \binom{m}{2}$ for all m : ruling out cancellation *among* the critical c_S , i.e. $G_{\binom{m}{2}} \neq 0$. The clean sufficient statement (verified $m \leq 4$) is that *every* critical survivor has $c_S = q^m/(q+1)^{2m}$ — all equal, hence positive, so $G_{\binom{m}{2}} = C_m q^m/(q+1)^{2m}$. This should follow from a telescoping of the gap–sandwich products (each survivor is a product of single–junction monomials $\beta\alpha^{k-2}$ and empty–gap G –factors, combinatorially constrained to a common value), or from Hoefsmit positivity ([hoefsmit-positivity]) forcing a common sign. The Catalan count C_m is a bonus, awaiting a ballot/Dyck bijection on the admissible (u_k, v_k) splittings with $u_k + v_k = k - 2$ (cf. [hook-support-lemma], [hook-ballot-dichotomy]). For $m \leq 4$ the order law is unconditional.

Toward general $\lambda = (\hat{\lambda}, 1^m)$. The rank–one collapse is special to hooks; off hooks $P_q^{(i)}$ has higher rank and c_S no longer factorises into scalars. The verified off–hook behaviour at $(2, 2, 1, 1, 1)$ suggests replacing the path–graph chip–firing by a firing process on the seminormal branching graph of $\hat{\lambda}$, with the same “order = minimal (-1) –insertion support, no cancellation” principle. That is the natural next target.